12.13 The Truel

WORKSHEET

The question

STEP II 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself In a two-person duel, Percy throws first. Either he hits at once, or both miss and the situation repeats exactly. Write an equation for Percy's probability of winning.

STAGE 2 Algernon's three options

Ask yourself In each case, who is left, and who throws next?

STAGE 3 Advice

Ask yourself If Algernon aims at Cuthbert, he might hit (probability $\frac{2}{5}$) or miss. What is his overall chance? Compare with deliberately missing.

Try this next

1. **Bertie and Cuthbert.** Verify that, with Algernon missing deliberately, Bertie wins with probability $\frac{135}{475}$ and Cuthbert with $\frac{114}{475}$.
2. **A better shot.** If Algernon hit with probability a instead of $\frac{2}{5}$, for which values of a is deliberately missing still his best first move?
3. **Should Bertie aim at Cuthbert?** Given that Algernon misses deliberately, is it in Bertie's interest to aim at Cuthbert rather than at Algernon? (Assume Cuthbert's behaviour is unchanged.)
4. **A duel with a fixed number of throws.** In a duel where each player has only two throws, find Percy's probability of winning, and compare with the unlimited duel.

Further reading

- **D. Marc Kilgour & Steven Brams, "The truel"**, *Mathematics Magazine* 70 (1997). An accessible survey of three-way duels.
- **Avinash Dixit & Barry Nalebuff, *Thinking Strategically*** (Norton). Game theory for a general reader, including the truel.
- **Ken Binmore, *Game Theory: A Very Short Introduction*** (Oxford University Press).

The full guided solution, and what the question leads to at university, is in Section 12.13:
https://beyondstep.keithmcnulty.org/ch12/truel.html